

Problem Set 1 – Diagnostic

Session 1 · Course overview and school math refresher

Part A · Algebraic care

1 EXPONENTS

Simplify $\frac{27^{2/3} \cdot 3^{-2}}{9^{1/2}}$, stating which exponent law you use at each step.

2 RADICALS

Decide whether $\sqrt{x^2} = x$ holds for all real x . If not, give the correct identity and a value of x that breaks the false one.

3 IDENTITIES

Prove that $(a + b)^2 - (a - b)^2 = 4ab$ for all real a, b . State explicitly why your argument covers *all* real a, b and not just the ones you tried.

4 COMPLETING THE SQUARE

Write $2x^2 - 12x + 23$ in the form $a(x - h)^2 + k$ and read off its minimum value and where it occurs.

Part B · Absolute value and inequalities

5 ABSOLUTE VALUE

Solve $|3x + 1| \geq 4$. Mark each step $\iff$ or $\implies$ and say why.

6 RATIONAL INEQUALITY

Find all real x with $\frac{2}{x-1} \leq 1$. Handle the sign of $x - 1$ explicitly; do not multiply through by an expression of unknown sign.

7 AM-GM

Show $x^2 + y^2 \geq 2xy$ for all real x, y , and determine exactly when equality holds.